

Dividing in a given ratio

A quantity can be divided into two, three, or more parts, according to a given ratio. This example shows how to divide 20 people into the ratios 2 : 3 and 6 : 3 : 1.



DIVIDING INTO A
TWO-PART RATIO

2 : 3

total number of parts in the ratio

$$2 + 3 = 5$$

number of parts in the ratio
total number of people

$$20 \div 5 = 4$$

2 in the ratio

$$2 \times 4 = 8$$

3 in the ratio

$$3 \times 4 = 12$$

12 people
represented by
3 in the ratio

8 people
represented by
2 in the ratio

These are the ratios to
divide the people into.

Add the different
parts of the ratio to
find the total parts.

Divide the number of people
by the parts of the ratio.

Multiply each part of
the ratio by this
quantity to find the size
of the groups the ratios
represent.

DIVIDING INTO A
THREE-PART RATIO

6 : 3 : 1

$$6 + 3 + 1 = 10$$

total number of people

$$20 \div 10 = 2$$

total number of
parts in the ratio

6 in the
ratio

$$6 \times 2 = 12$$

12 people
represented by
6 in the ratio

3 in the
ratio

$$3 \times 2 = 6$$

6 people
represented by
3 in the ratio

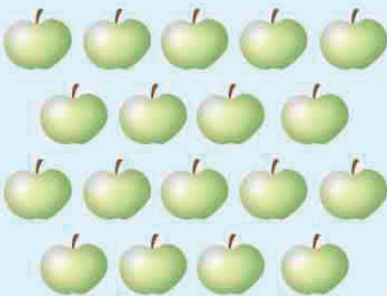
1 in the
ratio

$$1 \times 2 = 2$$

2 people
represented by
1 in the ratio

Proportional quantities

Proportion can be used to solve problems involving unknown quantities. For example, if 3 bags contain 18 apples, how many apples do 5 bags contain?



total number
of apples

$$18 \div 3 = 6$$

bags

apples
per bag

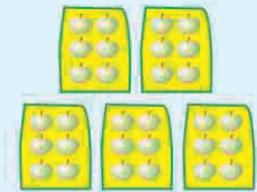


apples
per bag

number of bags

$$6 \times 5 = 30$$

total



There is a total of 18 apples in 3 bags.
Each bag contains the same number
of apples.

To find out how many apples there
are in 1 bag, divide the total number
of apples by the number of bags.

To find the number of apples
in 5 bags, multiply the number
of apples in 1 bag by 5.